

Dual Optimal Transport for Multi-Concept Composition: Structure Alignment and Texture Injection in Diffusion Models

Hao Fu¹ Tianyu Su¹ Meng Liu² Chenfang Yang³ Tian Gan¹

Abstract

Diffusion models have shown impressive capabilities in text-to-image synthesis. However, multi-concept personalized generation remains challenging, particularly in aligning multiple reference concepts while preserving fidelity. To address this, we propose a novel Sketch-to-Rendering framework that leverages *Dual Optimal Transport (OT)* for structure alignment and texture injection. Our approach consists of two key components: *Structure Sketching via Barycentric Soft-Transport*, which aligns concept geometry through relaxed barycentric feature transport, and *Texture Rendering via Geometry-Guided Transport*, which leverages low-frequency structure alignment to inject high-frequency texture details via OT-based residual transfer, thereby preserving texture fidelity without distorting structure. Extensive experiments demonstrate that our method significantly enhances both conceptual fidelity and visual quality. Ablation studies further validate the effectiveness of our optimal transport guidance and the decoupling of structure and texture during the generation process. Our code is available at <https://github.com/fuhao7i/OTComp>.

1. Introduction

The widespread availability of large-scale text-to-image (T2I) diffusion models (Rombach et al., 2022) has shifted the research frontier from generic image synthesis toward personalized customization. Beyond generating visually plausible images, users increasingly expect models to compose multiple *specific* concepts, such as “my pet dog”

¹School of Computer Science and Technology, Shandong University, Qingdao, China ²School of Computer and Artificial Intelligence, Shandong Jianzhu University, Jinan, China ³Haide College, Ocean University of China, Qingdao, China. Correspondence to: Tian Gan <gantian@sdu.edu.cn>.

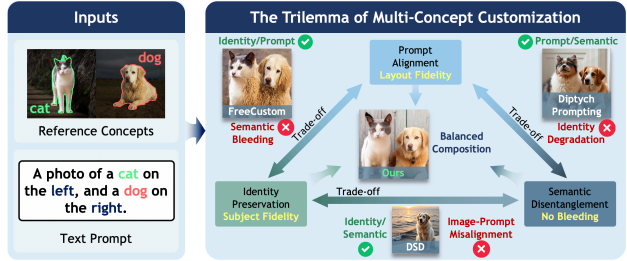


Figure 1. The Trilemma of Multi-Concept Customization. Multi-concept customization requires balancing identity fidelity, prompt-consistent layout, and semantic disentanglement. Existing methods often trade off one objective against another, whereas OTComp aims to achieve a more balanced composition across all three.

and “a particular plush toy”, into a single coherent scene. While single-concept personalization methods like Low-Rank Adaptation (LoRA) (Hu et al., 2022) and Dream-Booth (Ruiz et al., 2023) can faithfully capture individual identities, extending them to multi-concept composition remains challenging. Existing approaches often suffer from *concept bleeding*, where attributes leak across concepts, and *identity confusion*, where distinct subjects become entangled (Kumari et al., 2023; Gu et al., 2023).

At a deeper level, as shown in Figure 1, this challenge arises from a fundamental tension among three competing objectives: maintaining a prompt-consistent *global layout*, preserving fine-grained *identity fidelity* for each reference concept, and ensuring *semantic disentanglement* across concepts. A successful method must satisfy all three simultaneously. In practice, however, most existing approaches cast customization as a single, uniform control problem applied throughout the entire denoising trajectory. This stage-agnostic treatment inevitably leads to a zero-sum trade-off, *i.e.*, aggressively enforcing identity cues tends to override the prompt-consistent structure, while rigid layout control suppresses high-frequency identity details, with cross-concept interference further amplifying both failure modes.

Existing solutions primarily address this tension through *parameter-space merging* or *spatial-masking constraints*. Weight-merging approaches (Gu et al., 2023) assume that linearly combining fine-tuned parameters produces a seman-

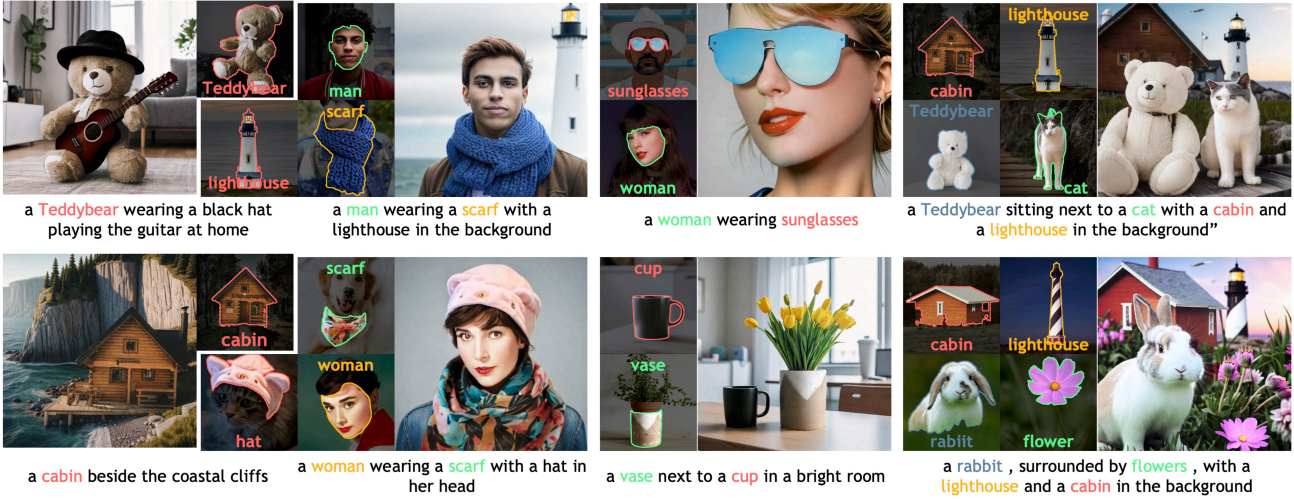


Figure 2. **High-Fidelity Multi-Concept Composition.** We present a training-free framework capable of synthesizing complex scenes. As illustrated, our approach achieves exceptional identity preservation and precise attribute binding, robustly handling challenging interactions (e.g., *playing guitar*) without semantic bleeding.

tic union of concepts. Yet, due to the highly non-convex nature of deep network parameter spaces, independently learned concepts often interfere and compete for shared representations (Jang et al., 2024). Conversely, spatial-masking methods (Lim et al., 2025; Avrahami et al., 2023) enforce hard spatial separation to isolate concepts. While effective at reducing interference, such rigid boundaries often suppress natural interactions such as lighting, shadows, and physical contact, and they become brittle when reliable layout priors are unavailable.

Crucially, these limitations stem from a deeper mismatch between *how diffusion models generate images* and *how multi-concept control is imposed*. Recent analyses show that diffusion follows a clear coarse-to-fine generative hierarchy: low-frequency global attributes (e.g., layout, pose, and spatial configuration) emerge early in the denoising process, while high-frequency details (e.g., texture, fine-grained identity) are resolved progressively in later stages (Wang & Pehlevan, 2026). Ignoring this hierarchy leads to systematic failure: imposing identity- or texture-level constraints too early disrupts global structure, whereas introducing structural constraints too late fails to prevent identity drift and concept entanglement. These observations naturally raise a key question: *How can we design a multi-concept personalization mechanism that respects the intrinsic coarse-to-fine hierarchy of diffusion models, establishing global structure before rendering fine-grained identity, while effectively preventing cross-concept interference?*

To answer this question, we propose OTComp, a training-free framework that formulates multi-concept customization as *transport-guided manifold traversal*. Instead of modifying model parameters, OTComp explicitly orchestrates

inference into three stages aligned with the diffusion hierarchy. (i) *Layout planning via diffusion prior*: we first sample a text-faithful *design blueprint* from the base T2I model to anchor global composition and extract per-concept spatial regions. (ii) *Structure sketching via barycentric soft-transport*: during early denoising, we guide concept geometry using a relaxed optimal transport objective that performs barycentric re-mapping in feature space, enabling non-rigid structural alignment without pixel-level copying. (iii) *Texture rendering via geometry-guided transport*: in later stages, we inject high-frequency identity details through a geometry-gated, mass-preserving Sinkhorn transport plan that transfers texture while preserving established structure. To further suppress semantic leakage, OTComp introduces a *concept-wise masked multi-reference attention* mechanism, which imposes a union-window connectivity constraint on the attention graph. This design allows localized interactions necessary for visual coherence while blocking long-range contamination between concepts and background regions. Our contributions are summarized as follows:

- We propose OTComp, a training-free framework that aligns multi-concept customization with diffusion’s coarse-to-fine dynamics via transport-based structure and texture control.
- We introduce a concept-wise masked multi-reference attention that enforces a union-window connectivity constraint to prevent concept bleeding while preserving localized interactions.
- Experiments and ablations show that OTComp improves both identity preservation and prompt alignment in challenging multi-concept composition scenarios.

2. Related Work

Multi-Concept Composition in Diffusion Models.

Diffusion-based personalization methods such as Textual Inversion (Gal et al., 2023), DreamBooth (Ruiz et al., 2023), and LoRA (Hu et al., 2022) have achieved impressive photorealism for single-concept learning. However, extending these approaches to multi-concept composition within a single coherent scene remains challenging. Existing solutions largely fall into two categories: parameter-space merging and attention-based spatial control.

Parameter-space merging methods, including Mix-of-Show (Gu et al., 2023) and recent LoRA-based weight merging strategies, attempt to combine independently learned concepts by linearly interpolating fine-tuned parameters. Due to the highly non-convex optimization landscape of deep diffusion models, such interpolation often causes gradients associated with dominant concepts to overwrite low-energy directions corresponding to more subtle identities, resulting in concept drift or spectral interference. Custom Diffusion (Kumari et al., 2023) mitigates this issue by introducing constrained optimization over shared parameters, but it requires computationally expensive joint training for each new concept combination, limiting scalability. In contrast, training-free attention control methods manipulate cross-attention maps to enforce spatial separation of concepts (Chefer et al., 2023; Phung et al., 2024; Avrahami et al., 2023). While effective for coarse object placement, rigid masking disrupts cross-token co-adaptation across attention heads, preventing the diffusion process from jointly modeling geometry-consistent illumination, shadows, and fine-grained interactions, which often leads to cut-and-paste artifacts and reduced realism. These limitations motivate approaches that operate directly on the feature manifold, enabling continuous concept interaction without relying on parameter fusion or hard spatial constraints.

Optimal Transport in Generative Modeling. Optimal Transport (OT) has emerged as a powerful tool for understanding and improving generative models. At the distributional level, Flow Matching (Lipman et al., 2023) and Rectified Flow (Liu et al., 2023) reformulate diffusion as an OT problem, learning straight-line probability paths that minimize transport cost. These approaches focus on global training objectives and the geometry of probability flows. At the representation level, OT has been widely applied to domain adaptation and few-shot generation (FAtlas et al., 2021; Turpault et al., 2019), with Sliced Wasserstein distances being successfully used for texture and style synthesis (Heitz et al., 2021). However, most existing methods apply OT as a global loss on pixel space or final-layer features, without explicitly accounting for the hierarchical representation structure and time-dependent denoising dynamics inherent in diffusion models. This gap highlights the need for OT for-

mulations that respect both multi-scale feature hierarchies and the progressive nature of diffusion-based generation.

Spectral Dynamics and Coarse-to-Fine Generation. Recent theoretical analyses show that diffusion models exhibit a pronounced spectral bias, prioritizing low-frequency structural components during high-noise stages and progressively refining high-frequency details as noise decreases (Choi et al., 2022; Wang & Pehlevan, 2026; Si et al., 2024). This coarse-to-fine behavior mirrors observations from human drawing and perception processes (Kozbelt, 2001). Despite this intrinsic temporal hierarchy, many personalization and composition methods inject high-frequency identity constraints, such as texture-level details, too early in the denoising process, interfering with global structure formation. Several works have explored frequency-aware guidance for style transfer (Mao et al., 2023) or spatial control via sketches (Voynov et al., 2023), but they lack a unified mechanism for coordinating structure and texture across multiple personalized concepts. These observations motivate a stage-aware perspective on multi-concept personalization, in which identity information is introduced in alignment with the natural spectral dynamics of diffusion, separating global structural sketching from fine-grained texture realization.

3. Method

In this section, we introduce OTComp, a training-free framework designed for high-fidelity multi-concept composition. Drawing inspiration from the spectral dynamics of diffusion physics and the coarse-to-fine nature of human artistic cognition, we propose to disentangle the customization process from the monolithic denoising trajectory. Specifically, the OTComp framework decomposes generation into three hierarchical stages: *Layout Planning* (Section 3.1), which anchors the global composition; *Structure Sketching* (Section 3.2), which aligns concept geometry via relaxed barycentric transport; and *Texture Rendering* (Section 3.3), which injects high-frequency details under low-frequency geometric guidance.

3.1. Layout Planning via Diffusion Prior

Given a prompt y and n reference concepts $\{x^c\}_{c=1}^n$, we first extract a *design blueprint* that captures the prompt-implied composition, *i.e.*, object presence, relative spatial arrangement, and global scene structure. Concretely, we sample a design image using a frozen, pretrained text-to-image (T2I) diffusion model ϵ_θ : $z_{\text{des}} \sim p_\theta(z \mid y)$. This step exploits the strong compositional prior of the base model to ensure prompt fidelity and coherent layout before introducing any personalization signals from the reference concepts.

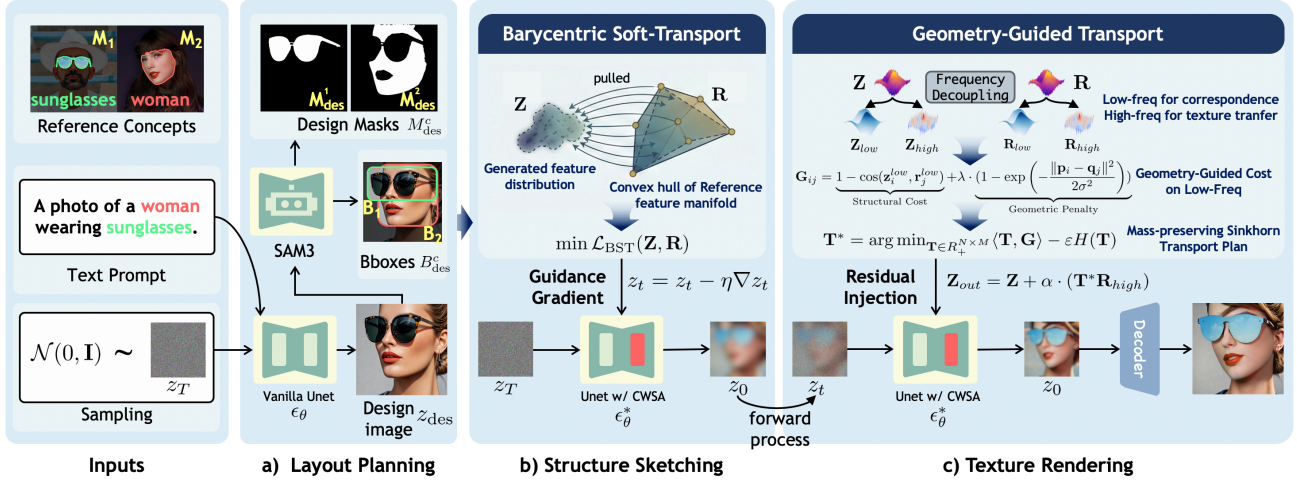


Figure 3. Overview of the proposed OTComp. OTComp decomposes training-free multi-concept composition into layout planning, structure sketching, and texture rendering. Barycentric Soft-Transport (BST) guides early structural alignment, Geometry-Guided Transport (GGT) injects late-stage texture details, and Concept-wise Masked Multi-Reference Attention (CWSA) constrains reference injection to mitigate cross-concept semantic leakage.

We then convert the design image into explicit spatial anchors by applying an off-the-shelf SAM3 model (Carion et al., 2025), yielding per-concept regions: $\{M_{\text{des}}^c, B_{\text{des}}^c\}_{c=1}^n$, where M_{des}^c denotes the mask and B_{des}^c is the bounding box associated with the c -th concept. These regions serve as *layout constraints* for subsequent concept injection.

Finally, each reference concept is spatially aligned to the design blueprint (e.g., via bounding box-guided warping), ensuring that concept features are compared and integrated under consistent scale and relative geometry.

3.2. Structure Sketching via Barycentric Soft-Transport

To inject the structural characteristics of reference concepts into the generation process without enforcing rigid, pixel-level correspondence, we propose *Barycentric Soft-Transport* (BST). Unlike classical optimal transport, which seeks a hard permutation, BST aligns the generated feature distribution with the *convex hull* of the reference features, enabling robust geometric interpolation.

Let $\mathbf{Z} \in \mathbb{R}^{N \times d}$ denote the set of feature tokens from the generated image (query) and $\mathbf{R} \in \mathbb{R}^{M \times d}$ be the set of feature tokens from the reference images (key). Both are normalized to the unit hypersphere $\mathbb{S}^{d-1}$. We define the pairwise cosine cost matrix $C \in \mathbb{R}^{N \times M}$ as $C_{ij} = 1 - \mathbf{z}_i^\top \mathbf{r}_j$. A simple hard-alignment baseline is to match each query token to its nearest reference token:

$$\mathcal{L}_{\text{Hard}} = \sum_{i=1}^N \min_j C_{ij} = \sum_{i=1}^N (1 - \max_j \mathbf{z}_i^\top \mathbf{r}_j). \quad (1)$$

While intuitive, this objective creates a loss landscape with

sharp discontinuities. The gradient $\nabla_{\mathbf{z}_i} \mathcal{L}_{\text{Hard}}$ depends solely on a single reference point $\mathbf{r}_{j^*}$. As $\mathbf{z}_i$ evolves during denoising, the index j^* can flip abruptly, causing gradient chatter and hindering the synthesis of geometrically coherent concepts.

To mitigate this, we introduce an entropic relaxation that maps each query token $\mathbf{z}_i$ to a local barycenter of the reference manifold, as shown in Figure 3 (b). We first compute a soft transport plan $\Pi \in \mathbb{R}^{N \times M}$ using a temperature-scaled Softmax over the top- k nearest neighbors $\mathcal{N}_k(i)$:

$$\pi_{ij} = \frac{\exp(-C_{ij}/\tau) \cdot \mathbb{I}_{j \in \mathcal{N}_k(i)}}{\sum_{l \in \mathcal{N}_k(i)} \exp(-C_{il}/\tau)}, \quad (2)$$

where τ is a temperature parameter controlling the smoothness of the transport. Instead of transporting mass to discrete points, we project the mass onto the Barycentric Target $\hat{\mathbf{z}}_i$, defined as the weighted average of reference tokens:

$$\hat{\mathbf{z}}_i = \text{Normalize} \left(\sum_{j=1}^M \pi_{ij} \mathbf{r}_j \right). \quad (3)$$

The final objective minimizes the divergence between the generated tokens and their barycentric projections within the union of concept masks $\mathcal{M}_{\text{union}}$:

$$\mathcal{L}_{\text{BST}} = \sum_{i \in \mathcal{M}_{\text{union}}} \|\mathbf{z}_i - \hat{\mathbf{z}}_i\|^2 \quad (4)$$

To enforce this structural constraint during inference, we treat $\mathcal{L}_{\text{BST}}$ as an energy potential. Following the standard guidance formulation (see Appendix A for derivation), at each denoising step t , we compute the gradient of the structural energy with respect to the latent $\mathbf{z}_t$ and perform a

guidance update:

$$z_t \leftarrow z_t - \eta_{struct} \cdot \nabla_{z_t} \mathcal{L}_{BST}(z_t), \quad (5)$$

where η_{struct} is a time-dependent scalar controlling the guidance strength. This step effectively pulls the noisy latent code towards the valid structural manifold of the reference concepts before the standard denoising step.

3.3. Texture Rendering via Geometry-Guided Transport

Standard softmax-based cross-attention often suffers from the “winner-takes-all” phenomenon, leading to mode collapse where dominant reference texture patches are repetitively tiled. To address this, we introduce *Geometry-Guided Transport (GGT)*. Building on the structural alignment established by the *Barycentric Soft-Transport* in the previous stage, GGT exploits the observation that semantic correspondence is most robust in the low-frequency domain (shapes and contours). We therefore propose a *spectral decomposition* strategy: low-frequency components are utilized to compute transport costs, explicitly guiding the injection of high-frequency texture details into geometrically coherent regions while enforcing mass conservation via the Sinkhorn-Knopp algorithm.

Formally, we decompose feature maps into low and high frequencies: $\mathbf{F}_{low} = \text{AvgPool}(\mathbf{F})$ and $\mathbf{F}_{high} = \mathbf{F} - \mathbf{F}_{low}$. Let $\mathbf{Z}_{low} \in \mathbb{R}^{N \times d}$ and $\mathbf{R}_{low} \in \mathbb{R}^{M \times d}$ denote the flattened low-frequency generated and reference features, respectively, with row vectors $\mathbf{z}_i^{low}$ and $\mathbf{r}_j^{low}$. The transport cost matrix $\mathbf{G} \in \mathbb{R}^{N \times M}$ is computed solely on the structural low-frequencies, modulated by a geometric penalty to enforce spatial locality:

$$\mathbf{G}_{ij} = \underbrace{1 - \cos(\mathbf{z}_i^{low}, \mathbf{r}_j^{low})}_{\text{Structural Cost}} + \lambda \cdot \underbrace{\left(1 - \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{q}_j\|^2}{2\sigma^2}\right)\right)}_{\text{Geometric Penalty}} \quad (6)$$

where $\mathbf{p}_i$ and $\mathbf{q}_j$ denote the normalized spatial coordinates of the generated and reference tokens, respectively.

Subsequently, to prevent texture repetition, we seek a balanced transport plan $\mathbf{T}^*$ that respects the conservation of texture mass. We formulate this as an entropically regularized optimal transport problem:

$$\begin{aligned} \mathbf{T}^* &= \arg \min_{\mathbf{T} \in \mathbb{R}_+^{N \times M}} \langle \mathbf{T}, \mathbf{G} \rangle - \varepsilon H(\mathbf{T}), \\ \text{s.t. } \mathbf{T} \mathbf{1}_M &= \mathbf{1}_N, \quad \mathbf{T}^\top \mathbf{1}_N = \frac{N}{M} \mathbf{1}_M, \end{aligned} \quad (7)$$

where $H(\mathbf{T})$ denotes the entropy regularizer. This optimization is solved efficiently via log-space Sinkhorn iterations. Finally, high-frequency texture details are injected into the generated features $\mathbf{Z}$ as a residual update:

$$\mathbf{Z}_{out} = \mathbf{Z} + \alpha \cdot (\mathbf{T}^* \mathbf{R}_{high}) \quad (8)$$

3.4. Concept-wise Masked Multi-Reference Attention

Beyond transport objectives, high-fidelity multi-concept composition crucially depends on *how* reference information is allowed to flow into the generation branch. A major failure mode in multi-subject customization is *semantic leakage*: reference attributes may bleed across concepts or contaminate the background, especially under occlusion and proximity. To address this, we introduce a concept-wise union-window masked multi-reference attention (CWSA) mechanism that serves as an *interaction backbone* for our framework.

We view self-attention as a message-passing graph over spatial tokens. Instead of permitting global interactions, we impose a *connectivity constraint*: only tokens inside a concept-specific window are allowed to aggregate information from the corresponding reference concept, while tokens outside the union of all concept windows remain self-contained. This design provides two complementary benefits: (i) *concept-wise isolation* prevents cross-concept interference by construction, and (ii) *background protection* blocks reference-to-background contamination by restricting reference access to semantic neighborhoods.

Let Q, K, V denote the query, key, and value tensors from the current generation branch, and $\{K_{ref}^c, V_{ref}^c\}_{c=1}^n$ denote key/value tensors extracted from the aligned reference branches. Let W^c be the window for concept c (obtained by expanding its mask) and let $W = \bigcup_{c=1}^n W^c$ be the union window. For a query token at spatial index i , the attention update is defined as

$$\text{Attn}(q_i) = \begin{cases} \text{softmax}\left(\frac{q_i [K; K_{ref}^c]^\top}{\sqrt{d}}\right) [V; V_{ref}^c], & i \in W^c, \\ \text{softmax}\left(\frac{q_i K^\top}{\sqrt{d}}\right) V, & i \notin W, \end{cases} \quad (9)$$

where d is the channel dimension. Eq. (9) formalizes a union-window gating rule: reference interactions are enabled *only* within concept neighborhoods, while the background (and irrelevant regions) is updated purely by self-attention. Importantly, because the reference branch is indexed by c , the interaction is *concept-wise* and does not mix reference concepts unless their windows overlap.

This attention constraint is orthogonal to the proposed transport objectives. Concept-wise union-window attention restricts *where* cross-reference messages can propagate, providing leakage-free feature aggregation. Our dual OT modules then specify *what* should be aligned and transferred within the allowed regions: barycentric transport guides early structural sketching, and geometry-gated Sinkhorn plans inject high-frequency textures in the rendering stage.

We implement Eq. (9) by rewriting the U-Net self-attention blocks to support concept-indexed reference aggregation under a union-window mask, with a stage-specific variant

Table 1. **Quantitative evaluation of customized multi-concept generation on MagicBench.** Best results are shown in bold. The second-best results are highlighted with underlines. FT denotes fine-tuning.

Method	Base Model	FT	Two concepts			Three concepts			Four concepts		
			CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINO $\uparrow$	CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINO $\uparrow$	CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINO $\uparrow$
Textual Inversion	SDXL	Yes	0.3303	0.5605	0.4542	0.3174	0.5298	0.4350	0.3223	0.4682	0.3732
DreamBooth	SDXL	Yes	0.3321	0.6937	0.5342	0.3416	0.6460	0.5150	0.3428	0.5781	0.4415
Custom Diffusion	SD2.1	Yes	0.3279	0.6128	0.4644	0.3460	0.6296	0.4689	0.3496	0.5490	0.4074
BLIP Diffusion	SD1.5	Yes	0.2741	0.6690	0.5609	0.2876	0.6025	0.4831	0.3044	0.5444	0.4357
IPAdapter	SD1.5	Yes	0.3007	0.6974	0.5644	0.3265	0.6352	0.4850	0.3498	0.5983	0.4512
IPAdapter+	SD1.5	Yes	0.3096	0.6892	0.5620	0.3258	0.6362	0.5170	0.3443	0.6112	0.4878
FreeCustom	SD1.5	No	0.2934	0.6450	0.5367	0.3139	0.5522	0.4195	0.3160	0.4568	0.3311
Emu2	MLLM	Yes	0.2861	0.5898	0.5006	0.2867	0.5040	0.4291	0.2907	0.4666	0.3851
Qwen2VL-Flux	FLUX.1	Yes	0.2774	0.6872	0.5015	0.2864	0.6150	0.4288	0.2998	0.5995	0.4171
Diptych Prompting	FLUX.1	No	<u>0.3503</u>	<u>0.7664</u>	<u>0.6388</u>	0.3521	0.6974	<u>0.5827</u>	0.3488	0.5964	0.5175
Diffusion Self-Dis.	FLUX.1	Yes	0.3358	0.6948	0.5605	0.3354	0.6645	0.5374	0.3545	0.6561	0.5423
OminiControl	FLUX.1	Yes	0.3434	0.7106	0.5666	<u>0.3532</u>	0.6836	0.5359	<u>0.3606</u>	0.6223	0.5068
Ours	SDXL	No	0.3528	0.8188	0.7455	0.3573	0.7794	0.7062	0.3828	0.7627	0.6844

in the early sketching phase.

4. Experiments

4.1. Experimental Settings

Implementation Details. We employ Stable Diffusion XL (SDXL) (Podell et al., 2024) as our backbone due to its superior high-resolution synthesis capabilities. To leverage robust priors in compositional generation, we initialize our model using the weights from IterComp (Zhang et al., 2025). This initialization provides a strong starting point for handling complex text prompts, spatial relationships, and attribute binding. All images are generated at a resolution of 768×768 . Inference is conducted on two NVIDIA RTX 4090 GPUs. After layout planning, the denoising process is divided into two guidance phases aligned with diffusion’s coarse-to-fine dynamics. 1) *Phase 1: Structure Sketching*. We apply the proposed Barycentric OT guidance during the early denoising timesteps $t \in [5, 20]$. For each guidance step, we perform $K = 5$ latent optimization iterations with a guidance scale of $\lambda_{struct} = 1.0$. This phase establishes the global layout without over-constraining fine details. 2) *Phase 2: Texture Rendering*. We inject high-frequency details using the Geometry-Gated Sinkhorn mechanism. This injection is selectively applied at timesteps $\mathcal{T}_{tex} = \{10, 11, 12\} \cup \{16, \dots, 21\}$. This specific interval was empirically determined to maximize texture fidelity while preserving the established structure. We replace the original self-attention modules in denoising U-Net blocks $\mathcal{B} = \{35, \dots, 69\}$ with *MRSA* during the structure sketching phase and *CWSA* during the texture rendering phase.

Datasets. Current evaluation protocols for diffusion-based personalization remain largely confined to simple, low-complexity scenarios. Standard benchmarks such as DreamBench (Ruiz et al., 2023) and CustomConcept101 (Kumari et al., 2023) primarily focus on single-concept customization, with limited scale and diversity (e.g., DreamBench includes only 30 concepts). Although FreeBench (Ding

et al., 2024) extends evaluation to multi-concept generation, it remains restricted by small-scale construction and simplistic prompt patterns. In existing benchmarks, prompts typically follow naive templates such as “subject + environment” (e.g., “a dog in the snow”) or “subject + accessory” (e.g., “a girl wearing a hat”), which fail to reflect real-world generation scenarios where multiple subjects, accessories, environments, and interactions must coexist coherently.

To address these limitations and enable rigorous evaluation under realistic compositional settings, we introduce *MagicBench*, a large-scale benchmark explicitly designed to stress-test multi-concept personalization. *MagicBench* comprises 100 reference concepts spanning 20 semantic categories, hierarchically covering animate subjects, rigid accessories, and spatial environments. Rather than relying on attribute-level prompt composition or manually enumerated simple templates, *MagicBench* is constructed from holistic, interaction-rich scene descriptions. We first use GPT-5.2 to synthesize candidate templates that jointly specify subjects, accessories, environments, and actions within a single coherent prompt. To ensure semantic plausibility, we manually filter the generated templates to remove unnatural or logically invalid compositions. The remaining templates are then systematically populated with reference categories, resulting in four progressive subsets corresponding to one-, two-, three-, and four-concept compositions, each containing 100 curated prompts. By increasing the number of interacting concepts while preserving natural scene structure, *MagicBench* provides a controlled testbed for evaluating semantic disentanglement, layout planning, and cross-concept interaction modeling. We report comparisons on both the established FreeBench and the proposed *MagicBench*.

Evaluation Metrics. Similar to FreeCustom (Ding et al., 2024), we assess image similarity using DINOv3 (Siméoni et al., 2025) and CLIP-I (Radford et al., 2021), and image-text consistency using CLIP-T.

Table 2. Quantitative evaluation of customized single-concept generation on MagicBench. CLIP-T measures prompt fidelity. CLIP-I and DINO measure image fidelity.

Method	Venue	CLIP-T↑	CLIP-I↑	DINO↑
Textual Inversion	ICLR'23	0.3379	0.7121	0.5937
DreamBooth	CVPR'23	0.3449	0.7537	0.6394
CustomDiffusion	CVPR'23	0.3242	0.5852	0.4400
BLIP Diffusion	NeurIPS'23	0.2723	0.8710	0.8186
IPAdapter	arXiv'23	0.2845	<u>0.8526</u>	0.7705
IPAdapter+	arXiv'23	0.2951	0.8471	0.7738
FreeCustom	CVPR'24	0.2935	0.8420	0.7609
Emu2	CVPR'24	0.3136	0.7935	0.7362
Qwen2VL-Flux	Report'24	0.2522	0.7319	0.5885
Diptych Prompting	CVPR'25	<u>0.3442</u>	0.7827	0.7178
Diffusion Self-Dis.	CVPR'25	0.3413	0.7762	0.7174
OminiControl	ICCV'25	0.3403	0.7447	0.6797
Ours	This work	0.3409	0.8356	<u>0.8174</u>



Figure 4. Comparisons of single- and multi-concept composition. OTComp preserves reference identities and attribute binding more reliably under challenging subject-accessory interactions.

Compared Methods. To rigorously evaluate the efficacy of our method in both single-concept and multi-concept customization, we benchmark against a comprehensive suite of state-of-the-art approaches, covering three distinct paradigms: 1) *Training-Free Methods*: We compare with FreeCustom (Ding et al., 2024) and Diptych Prompting (Shin et al., 2025), which manipulate attention or guidance without parameter updates. 2) *Tuning-Based Approaches*: We compare with DreamBooth (Ruiz et al., 2023) and Diffusion Self-Distillation (Cai et al., 2025), which customize generation through subject-specific fine-tuning or self-distillation. 3) *Large-Scale Pre-trained Models*: We also evaluate against foundation models with distinct architectures, including the adapter-based OminiControl (Tan et al., 2025) and the large multimodal model Emu2 (Sun et al., 2024).

4.2. Main Results

Quantitative Comparison. As detailed in Tables 2, 3, and 1, our method achieves state-of-the-art performance across both MagicBench and FreeBench. We consistently outperform strong baselines, including the FLUX.1-based OminiControl and optimization-based DreamBooth, particularly in identity fidelity metrics (CLIP-I and DINO), verifying the efficacy of our spectral decoupling. Crucially, Table 1 highlights our scalability: while competitor performance

Table 3. Quantitative evaluation of customized multi-concept generation on FreeBench. CLIP-T measures prompt fidelity. CLIP-I and DINO measure image fidelity.

Method	Base Model	CLIP-T↑	CLIP-I↑	DINO↑
Textual Inversion	SDXL	0.2822	0.5016	0.3733
DreamBooth	SDXL	0.3232	0.7118	0.5458
CustomDiffusion	SD2.1	0.3151	0.6732	0.4736
BLIP Diffusion	SD1.5	0.2566	0.6538	0.5198
IPAdapter	SD1.5	0.3129	<u>0.7517</u>	0.5702
IPAdapter+	SD1.5	0.3090	0.6878	0.5476
FreeCustom	SD1.5	0.3123	0.6643	0.5652
Emu2	MLLM	0.2777	0.5983	0.5004
Qwen2VL-Flux	FLUX.1	0.3012	0.6864	0.4648
Diptych Prompting	FLUX.1	0.3314	0.7265	<u>0.5811</u>
Diffusion Self-Dis.	FLUX.1	0.3087	0.6557	0.5160
OminiControl	FLUX.1	0.3194	0.7049	0.5324
Ours	SDXL	<u>0.3247</u>	0.7821	0.6747

degrades significantly as the number of concepts increases from 2 to 4, our approach maintains robust fidelity, widening the performance gap in complex compositional settings.

Qualitative Comparison. Visual comparisons in Figure 4 corroborate these metrics. While methods like Diptych Prompting and FreeCustom often suffer from *identity washout* (e.g., losing the specific reflective texture of the sunglasses) or *layout collapse* during interactions, our method successfully synthesizes coherent scenes with precise attribute binding and high-fidelity identity retention, even for challenging subject-accessory combinations.

Human Preference Evaluation. We further conduct a blind human preference study to assess whether the quantitative improvements are consistent with human judgments. The study includes 12 multi-concept test cases and five methods: OTComp, FreeCustom (Ding et al., 2024), Diptych Prompting (Shin et al., 2025), Diffusion Self-Distillation (Cai et al., 2025), and OminiControl (Tan et al., 2025). For each test case, participants are shown anonymized and randomly ordered results from all methods and are asked to rank them along three axes: identity fidelity, prompt alignment, and semantic disentanglement. We collect responses from 17 participants, yielding 204 rankings for each evaluation axis. Table 4 reports the top-1 preference rate, i.e., the percentage of rankings in which each method is selected as the best among the five candidates. OTComp obtains the highest preference rate on all three axes, with 74.51% for identity fidelity, 69.61% for prompt alignment, and 65.20% for semantic disentanglement. These results indicate that the improvements observed in automatic metrics are also reflected in human preference judgments.

4.3. Ablation Study

On OTComp Pipeline. To validate our core insight that multi-concept customization should be decoupled into *Planning*, *Sketching*, and *Rendering*, we visualize the generation trajectory in Figure 5.

Table 4. Blind human preference evaluation. We report the top-1 preference rate (%), defined as the percentage of rankings in which a method is selected as the best among five candidates. Results are computed over 204 rankings per axis from 17 participants and 12 multi-concept test cases.

Evaluation Axis	FreeCustom	Diptych	Diffusion Self-Dis.	OminiControl	OTComp
Identity fidelity	9.31%	4.90%	4.90%	6.37%	74.51%
Prompt alignment	2.45%	16.18%	7.35%	4.41%	69.61%
Semantic disentanglement	4.90%	11.76%	7.35%	10.78%	65.20%



Figure 5. Progressive ablation of the proposed OTComp pipeline.

Stage 1: Semantic Planning (Layout). As shown in the second column, relying solely on the T2I model’s layout priors allows us to generate a globally coherent scene with correct spatial relationships (e.g., a woman wearing a hat, a rabbit picking flowers). However, without identity guidance, the subjects remain generic approximations of the text prompt.

Stage 2: Structure Sketching (BST). Upon introducing the *Barycentric Soft-Transport*, we observe a critical transformation in the third column. The generated concepts align with the reference in terms of low-frequency attributes—such as color, shape, and pose—while preserving the original layout. Crucially, consistent with our spectral decoupling hypothesis, high-frequency details are notably absent. For instance, in the first row, the hat successfully morphs into a pink shape with animal ears, yet lacks the specific “pig face” texture (eyes and nose). This validates that BST effectively transports the *structural hull* of the reference without forcing premature texture injection that could disrupt the layout.

Stage 3: Texture Rendering (GGT). In the final stage (last column), the *Geometry-Guided Transport* injects the remaining high-frequency residuals. We observe the emergence of fine-grained identity details, such as the knitting texture and



Figure 6. Visual ablation of Structure Sketching via BST.

the facial features of the pig hat, and the specific fur pattern of the rabbit. This progressive evolution demonstrates that our method achieves a harmonious balance: leveraging the base model for *Planning*, BST for *Structure*, and GGT for *Texture*, thereby solving the trilemma of layout, identity, and disentanglement.

On Barycentric Soft-Transport. To validate the design choices of our *Barycentric Soft-Transport*, we conduct ablations across three dimensions: optimization mechanism, geometric flexibility, and temporal dynamics.

Optimization Mechanism. We compare our projection-based guidance against a naive baseline that directly minimizes the Sinkhorn loss ($\nabla_z \mathcal{L}_{OT}$). As shown in Figure 6(a), direct optimization yields structural instability and artifacts due to sparse gradients. In contrast, our barycentric approach serves as a manifold projection, ensuring the guided features remain within the valid semantic hull of the reference, resulting in smoother and more coherent structures.

Geometric Flexibility. We compare with a rigid feature matching baseline (Spatial MSE). Figure 6(b) demonstrates that rigid alignment fails when the target pose differs from the reference (e.g., “standing” vs. “sitting”), causing severe distortion. Our OT-based transport allows for geometric deformation, successfully preserving identity while adhering to the layout dictated by the text prompt.

Temporal Dynamics. Figure 6(c) shows that applying structural guidance in later timesteps (e.g., $t < 10$) leads to over-smoothing and texture loss. This confirms our spectral decoupling hypothesis: structural constraints must be imposed early ($t \in [5, 20]$) to align with the low-frequency formation phase of the diffusion process.



Figure 7. Visual ablation of attention mechanisms. **Yellow boxes** indicate semantic leakage (concept bleeding), **red boxes** denote identity degradation, and **green boxes** represent successful disentanglement with high fidelity.

On Attention Mechanisms. We evaluate the effectiveness of our proposed attention mechanism against the standard Multi-Reference Self-Attention (MRSA) used in FreeCustom (Ding et al., 2024) and a naive CMSA baseline. Visual comparisons are presented in Figure 7.

As shown in the first column (SDXL+MRSA), the standard MRSA allows unrestricted attention across all reference concepts. This lack of spatial constraint leads to severe *semantic leakage* (highlighted in yellow). In the top row, the texture of the reference pink hat bleeds into the scarf region. Similarly, in the bottom row, the identity of the reference grey cat contaminates the face of the auxiliary yellow cat. This confirms that without explicit geometric gating, global attention mechanisms struggle to disentangle co-occurring semantic features.

Our full approach resolves this trade-off through a two-stage synergy. First, comparing *Ours+MRSA* with *SDXL+MRSA*, we observe that our Stage 1 (Structure Sketching via Barycentric Soft-Transport) significantly alleviates semantic leakage even when using the baseline attention. This is because the optimal transport guidance pre-aligns the semantic distribution of the generated latent with the reference, reducing the ambiguity that causes attention misalignment. Finally, by integrating CWSA (*Ours+CWSA*), we achieve the optimal balance.

As shown in the last column (green boxes), CWSA spatially gates reference aggregation within the target regions, allowing identity-specific features to be injected without contaminating nearby concepts or background regions. It successfully injects the specific identity of the pink hat and grey cat without corrupting the surrounding context (scarf and yellow cat), demonstrating superior semantic disentanglement and fidelity.

5. Limitations

The effectiveness of OTComp depends on the reliability of the initial layout and segmentation cues. Our Barycentric Soft-Transport provides a spatially relaxed alignment mechanism and is robust to moderate mask noise, boundary dilation, and small localization errors. However, it cannot fully correct upstream failures. When the base model omits a concept, generates an implausible spatial arrangement, or when the segmentation module fails to separate heavily occluded objects, the resulting transport plan may propagate these errors by assigning features to incorrect or entangled regions. This can lead to texture leakage, incomplete concept preservation, or residual semantic entanglement. Future work may address these limitations by coupling transport with iterative layout verification and segmentation-aware feedback.

6. Conclusion

In this work, we have dismantled the long-standing trilemma in multi-concept customization by rethinking the generation process through a *spectral lens*. We demonstrate that the conflict between layout fidelity and identity preservation is not an inherent limitation of diffusion models, but rather an artifact of enforcing uniform constraints across disjoint frequency bands. By introducing a training-free framework that decouples customization into *Structure Sketching* (via Barycentric Soft-Transport) and *Texture Rendering* (via Geometry-Guided Transport), we achieve a harmonious balance that respects the intrinsic coarse-to-fine dynamics of the denoising process.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China, No.:62376140, and the Special Fund for Taishan Scholar Project of Shandong Province.

Impact Statement

This work advances controllable and personalized image generation by enabling training-free composition of multiple reference concepts into coherent scenes. The proposed method may benefit creative design, visual prototyping, education, and research on controllable generative models. At the same time, improved reference-based personalization may facilitate misleading or unauthorized depictions of identifiable individuals, copyrighted characters, brands, or other proprietary concepts. We therefore encourage responsible use with appropriate rights or consent for reference images, disclosure of synthetic content in sensitive settings, and safeguards such as provenance tracking, watermarking, and misuse detection prior to deployment.

References

- Avrahami, O., Hayes, T., Gafni, O., Gupta, S., Taigman, Y., Parikh, D., Lischinski, D., Fried, O., and Yin, X. Spatext: Spatio-textual representation for controllable image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18370–18380, 2023.
- Bansal, A., Chu, H.-M., Schwarzschild, A., Sengupta, S., Goldblum, M., Geiping, J., and Goldstein, T. Universal guidance for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 843–852, 2023.
- Cai, S., Chan, E. R., Zhang, Y., Guibas, L., Wu, J., and Wetzstein, G. Diffusion self-distillation for zero-shot customized image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18434–18443, 2025.
- Carion, N., Gustafson, L., Hu, Y.-T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K. V., Khedr, H., Huang, A., et al. SAM 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.
- Chefer, H., Alaluf, Y., Vinker, Y., Wolf, L., and Cohen-Or, D. Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. *ACM Transactions on Graphics (TOG)*, 42(4):1–10, 2023.
- Choi, J., Lee, J., Shin, C., Kim, S., Kim, H., and Yoon, S. Perception prioritized training of diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11472–11481, 2022.
- Ding, G., Zhao, C., Wang, W., Yang, Z., Liu, Z., Chen, H., and Shen, C. Freecustom: Tuning-free customized image generation for multi-concept composition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9089–9098, 2024.
- Fatras, K., Séjourné, T., Flamary, R., and Courty, N. Unbalanced minibatch optimal transport; applications to domain adaptation. In *International Conference on Machine Learning*, pp. 3186–3197. PMLR, 2021.
- Gal, R., Alaluf, Y., Atzmon, Y., Patashnik, O., Bermano, A. H., Chechik, G., and Cohen-Or, D. An image is worth one word: Personalizing text-to-image generation using textual inversion. In *The Eleventh International Conference on Learning Representations*, 2023.
- Geng, D. and Owens, A. Motion guidance: Diffusion-based image editing with differentiable motion estimators. In *The Twelfth International Conference on Learning Representations*, 2024.
- Gu, Y., Wang, X., Wu, J. Z., Shi, Y., Chen, Y., Fan, Z., Xiao, W., Zhao, R., Chang, S., Wu, W., et al. Mix-of-Show: Decentralized low-rank adaptation for multi-concept customization of diffusion models. *Advances in Neural Information Processing Systems*, 36:15890–15902, 2023.
- Heitz, E., Vanhoey, K., Chambon, T., and Belcour, L. A sliced wasserstein loss for neural texture synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9412–9420, 2021.
- Hu, E. J., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al. LoRA: Low-rank adaptation of large language models. In *The Tenth International Conference on Learning Representations*, 2022.
- Jang, S., Jo, J., Lee, K., and Hwang, S. J. Identity decoupling for multi-subject personalization of text-to-image models. *Advances in Neural Information Processing Systems*, 37:100895–100937, 2024.
- Kozbelt, A. Artists as experts in visual cognition. *Visual cognition*, 8(6):705–723, 2001.
- Kumari, N., Zhang, B., Zhang, R., Shechtman, E., and Zhu, J.-Y. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1931–1941, 2023.
- Lim, H., Won, Y., Seo, J., and Park, G.-M. ConceptSplit: Decoupled multi-concept personalization of diffusion models via token-wise adaptation and attention disentanglement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 18421–18430, 2025.
- Lipman, Y., Chen, R. T., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023.
- Liu, X., Gong, C., and Liu, Q. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *The Eleventh International Conference on Learning Representations*, 2023.
- Lv, H., Xiao, J., and Li, L. Pick-and-draw: Training-free semantic guidance for text-to-image personalization. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 10535–10543, 2024.
- Mao, J., Wang, X., and Aizawa, K. Guided image synthesis via initial image editing in diffusion model. In *Proceedings of the 31st ACM International Conference on Multimedia*, pp. 5321–5329, 2023.

- Phung, Q., Ge, S., and Huang, J.-B. Grounded text-to-image synthesis with attention refocusing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 7932–7942, 2024.
- Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., and Rombach, R. SDXL: Improving latent diffusion models for high-resolution image synthesis. In *The Twelfth International Conference on Learning Representations*, 2024.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pp. 8748–8763. PMLR, 2021.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695, 2022.
- Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., and Aberman, K. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22500–22510, 2023.
- Shin, C., Choi, J., Kim, H., and Yoon, S. Large-scale text-to-image model with inpainting is a zero-shot subject-driven image generator. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7986–7996, 2025.
- Si, C., Huang, Z., Jiang, Y., and Liu, Z. FreeU: Free lunch in diffusion u-net. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4733–4743, 2024.
- Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., Massa, F., Haziza, D., Wehrstedt, L., Wang, J., Darcet, T., Moutakanni, T., Sentana, L., Roberts, C., Vedaldi, A., Tolan, J., Brandt, J., Couprie, C., Mairal, J., Jégou, H., Labatut, P., and Bojanowski, P. DINOv3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Sun, Q., Cui, Y., Zhang, X., Zhang, F., Yu, Q., Wang, Y., Rao, Y., Liu, J., Huang, T., and Wang, X. Generative multimodal models are in-context learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14398–14409, 2024.
- Tan, Z., Liu, S., Yang, X., Xue, Q., and Wang, X. OminiControl: Minimal and universal control for diffusion transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 14940–14950, 2025.
- Turpault, N., Serizel, R., and Vincent, E. Semi-supervised triplet loss based learning of ambient audio embeddings. In *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 760–764. IEEE, 2019.
- Voynov, A., Aberman, K., and Cohen-Or, D. Sketch-guided text-to-image diffusion models. In *ACM SIGGRAPH 2023 conference proceedings*, pp. 1–11, 2023.
- Wang, B. and Pehlevan, C. An analytical theory of spectral bias in the learning dynamics of diffusion models. *Advances in Neural Information Processing Systems*, 38: 95865–95963, 2026.
- Wang, Z., Peng, D., Chen, F., Yang, Y., and Lei, Y. Training-free dense-aligned diffusion guidance for modular conditional image synthesis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 13135–13145, 2025.
- Zhang, X., Yang, L., Li, G., Cai, Y., Xie, J., Tang, Y., Yang, Y., Wang, M., and Cui, B. IterComp: Iterative composition-aware feedback learning from model gallery for text-to-image generation. In *The Thirteenth International Conference on Learning Representations*, 2025.

A. Preliminaries

Latent Diffusion Models. We build upon the Latent Diffusion Model (LDM) framework (Rombach et al., 2022), which performs diffusion in a learned latent space to generate an image conditioned on a text prompt p . Given an image x_0 , a pretrained autoencoder maps it to a low-dimensional latent representation $z_0 = E(x_0)$, and reconstructs it via $x_0 \approx D(z_0)$, where E and D denote the encoder and decoder, respectively. Diffusion is then applied to z_0 by progressively adding Gaussian noise to obtain z_t at timestep t . For text conditioning, the prompt p is encoded as $y = \phi(p)$ using a pretrained text encoder ϕ (Radford et al., 2021). The denoising network ϵ_θ is trained to predict the injected noise ϵ from the noised latent z_t , conditioned on t and y , by minimizing

$$\min_{\theta} \mathbb{E}_{z_0, \epsilon, t} \|\epsilon - \epsilon_\theta(z_t, t, y)\|_2^2. \quad (10)$$

Optimal Transport for Concept Alignment. Optimal transport (OT) provides a principled way to *match* two distributions by moving “mass” from one to the other with minimum total cost. In concept-preserving generation, OT (Lv et al., 2024) is commonly used to align the *generated* concept region with a *reference* concept in a learned feature space, thereby encouraging identity consistency. Concretely, for concept c , let $S^c = \{s_i^c\}_{i=1}^{N_c}$ and $T^c = \{t_j^c\}_{j=1}^{M_c}$ denote feature tokens extracted from the generated region and the reference region, respectively. A typical choice of the transport cost is a pairwise cost matrix $C^c \in \mathbb{R}^{N_c \times M_c}$, e.g., cosine distance:

$$C_{ij}^c = 1 - \frac{\langle s_i^c, t_j^c \rangle}{\|s_i^c\|_2 \|t_j^c\|_2}. \quad (11)$$

OT then finds a transport plan π^c that minimizes the expected matching cost under prescribed marginals:

$$\mathcal{E}_{\text{OT}}^c = \min_{\pi^c \in \Pi(a^c, b^c)} \langle \pi^c, C^c \rangle + \varepsilon \Omega(\pi^c), \quad (12)$$

where a^c and b^c are marginal distributions, $\Pi(a^c, b^c) = \{\pi \in \mathbb{R}_+^{N_c \times M_c} \mid \pi \mathbf{1} = a^c, \pi^\top \mathbf{1} = b^c\}$, and $\Omega(\cdot)$ is a differentiable regularizer (e.g., entropic regularization), which yields an efficient solver such as Sinkhorn iterations. For multiple concepts, the OT energies are typically combined as $\mathcal{E}_{\text{OT}} = \sum_{c=1}^n \lambda_c \mathcal{E}_{\text{OT}}^c$, with weights λ_c controlling the relative importance of concepts.

Backward Guidance for Customization. Backward guidance (Bansal et al., 2023; Geng & Owens, 2024; Wang et al., 2025) is a general sampling approach for diffusion models that steers generation toward additional constraints without retraining the denoiser. It targets sampling from a constraint-conditioned distribution, e.g., $p(z \mid y, \{x^c\}_{c=1}^n)$, where z denotes the diffusion latent and $y = \phi(p)$ is the text condition. The key idea is to specify an *optimizable energy* $\mathcal{E}(z_t)$ encoding the desired constraint (e.g., an optimal transport energy $\mathcal{E}_{\text{OT}}$), and to bias the reverse process toward reference-consistent concepts by taking a gradient step on z_t at each denoising timestep:

$$z_t \leftarrow z_t - \eta \nabla_{z_t} \mathcal{E}_{\text{OT}}(z_t), \quad (13)$$

where η controls the guidance strength.

B. From Discrete Points to Continuous Manifolds: Why Is This Superior to Naive EMD for Generation?

We provide additional details on the optimization behavior of the proposed Barycentric Soft-Transport (BST). Throughout this section, we use π_{ij} to denote the soft transport weight from the generated token $\mathbf{z}_i$ to the reference token $\mathbf{r}_j$. Given the cosine cost $C_{ij} = 1 - \mathbf{z}_i^\top \mathbf{r}_j$, the transport weights are computed as

$$\pi_{ij} = \frac{\exp(-C_{ij}/\tau) \mathbb{I}_{\{j \in \mathcal{N}_k(i)\}}}{\sum_{\ell \in \mathcal{N}_k(i)} \exp(-C_{i\ell}/\tau)}, \quad (14)$$

where $\mathcal{N}_k(i)$ denotes the top- k nearest reference tokens of $\mathbf{z}_i$, and τ controls the smoothness of the local transport plan. The barycentric target of $\mathbf{z}_i$ is then defined as

$$\hat{\mathbf{z}}_i = \text{Normalize} \left(\sum_{j=1}^M \pi_{ij} \mathbf{r}_j \right). \quad (15)$$

The BST objective used for structural guidance is

$$\mathcal{L}_{\text{BST}} = \sum_{i \in \mathcal{M}_{\text{union}}} \|\mathbf{z}_i - \hat{\mathbf{z}}_i\|_2^2. \quad (16)$$

Gradient approximation. In practice, we stop gradients through the transport weights π_{ij} when computing the guidance direction. This approximation avoids backpropagating through the soft assignment process and yields a stable projection-style update toward the local barycentric target. Formally, by the chain rule, the full gradient can be decomposed as

$$\nabla_{\mathbf{z}} \mathcal{L}_{\text{BST}} = \frac{\partial \mathcal{L}_{\text{BST}}}{\partial \mathbf{z}} + \left(\frac{\partial \pi_{\tau}}{\partial \mathbf{z}} \right)^{\top} \frac{\partial \mathcal{L}_{\text{BST}}}{\partial \pi_{\tau}}, \quad (17)$$

where π_{τ} denotes the temperature-dependent transport plan. As $\tau \rightarrow 0$, π_{τ} approaches a hard local assignment. Away from assignment boundaries, this limiting assignment is locally constant with respect to $\mathbf{z}$, and therefore its Jacobian vanishes almost everywhere:

$$\frac{\partial \pi_0}{\partial \mathbf{z}} = \mathbf{0} \quad \text{a.e.} \quad (18)$$

Under this small-temperature approximation, the second term in the chain-rule decomposition becomes negligible, leading to the detached-gradient update

$$\nabla_{\mathbf{z}_i} \mathcal{L}_{\text{BST}} \approx 2(\mathbf{z}_i - \hat{\mathbf{z}}_i). \quad (19)$$

Thus, the corresponding negative gradient direction pulls each generated token toward its local barycentric target:

$$-\nabla_{\mathbf{z}_i} \mathcal{L}_{\text{BST}} \propto \hat{\mathbf{z}}_i - \mathbf{z}_i. \quad (20)$$

This formulation can be interpreted as a smooth manifold projection: instead of forcing $\mathbf{z}_i$ to copy a single nearest reference token, BST moves it toward a convex combination of locally compatible reference tokens. This improves geometric stability during early denoising and avoids the numerical instability associated with unrolling the assignment or transport computation through multiple iterations.

Implementation Details: To handle multiple concepts $C = \{c_1, \dots, c_n\}$ without gradient conflict, we employ a Split-Mem strategy. We construct separate mini-batches $\{\mathbf{z}_{\text{base}}, \mathbf{z}_{\text{ref}}^{(c)}, \mathbf{z}_{\text{gen}}\}$ for each concept c . Furthermore, to handle occlusions and loose structural alignments in the early diffusion stage ($t \in$), we compute the loss only within a dynamic Union Mask:

$$M_{\text{union}}^{(c)} = M_{\text{pred}}^{(c)} \cup M_{\text{ref}}^{(c)},$$

where $M_{\text{pred}}^{(c)}$ is the attention-derived mask of the generated concept and $M_{\text{ref}}^{(c)}$ is the aligned design mask. This spatial relaxation ensures that the OT guidance is applied robustly even when the generated object’s shape slightly deviates from the initial layout prior.

C. Theoretical Analysis: Why Sinkhorn?

We provide a theoretical justification for replacing Softmax with Sinkhorn in Stage 2 from the perspective of balanced reference-token usage.

Proposition C.1. *Let $P \in \mathbb{R}_+^{N \times M}$ be the row-wise Softmax attention matrix and $T \in \mathbb{R}_+^{N \times M}$ be the Sinkhorn transport plan. Define the reference-token usage distributions as $u_j^{\text{soft}} = \frac{1}{N} \sum_i P_{ij}$ and $u_j^{\text{sink}} = \frac{1}{N} \sum_i T_{ij}$. If T satisfies the uniform column marginal $T^{\top} \mathbf{1}_N = \frac{N}{M} \mathbf{1}_M$, then*

$$\mathcal{H}(u^{\text{sink}}) \geq \mathcal{H}(u^{\text{soft}}), \quad (21)$$

where $\mathcal{H}(u) = -\sum_j u_j \log u_j$. Equality holds when u^{soft} is also uniform.

Proof Sketch. Softmax normalizes only row-wise, and therefore does not constrain how frequently each reference token is used across query locations. This can cause a small number of high-similarity reference tokens to dominate the injected features. In contrast, Sinkhorn additionally enforces a column marginal, yielding $u_j^{\text{sink}} = 1/M$ for all j . Since the uniform distribution maximizes entropy on the probability simplex, $\mathcal{H}(u^{\text{sink}}) = \log M \geq \mathcal{H}(u^{\text{soft}})$. This balanced usage reduces repetitive texture injection and encourages more reference details to participate in high-frequency residual transfer.